

Haibo Liu

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📍 School of Engineering, Westlake University, Hangzhou 310030, P.R. China.

Education

Tsinghua University	Beijing, China
Ph.D. in Mathematics. Advised by Prof. Junqing Chen	2019 / 8-2025 / 6
Lanzhou University	Lanzhou, China
B.S. in Mathematics and Applied Mathematics	2015 / 8-2019 / 6
Georgia State University	Atlanta, USA
Exchange in the Department of Mathematics and Statistics	2017 / 6-2017 / 9

Employment

Westlake University	Hangzhou, China
Research Assistant, with Prof. Xiangru Huang	2025 / 7-now

Research Interests

My research interests cover Computer Vision, AI4S, Physics-informed Machine Learning. More specifically:

- Dynamic 3D Reconstruction and Generation,
- Inverse Problems(Shape Optimization),
- 3D Medical Image Reconstruction,
- ...

Publications

- Implicit Neural Shape Optimization for 3D High-Contrast Electrical Impedance Tomography, (with Junqing Chen), arXiv preprint, (2025).
- Solving inverse obstacle scattering problem with latent surface representations, (with Junqing Chen and Bangti Jin), Inverse Problems, (2024).
- An Alternating Direction Method of Multipliers for Inverse Lithography Problem, (with Junqing Chen), Numerical Mathematics: Theory, Methods and Applications,16 (2023).

Awards and Honors

- **National Endeavor Scholarship** Lanzhou University of China 2016
- **The 9th National College Math Competition National Second Prize** China 2018

Scientific Events

- Co-organizer:“**0-1 Seminar**”, Tsinghua University July 2022

Teaching at Tsinghua University

- Associate teaching experience: Probability theory and mathematical statistics autumn 2020
- Associate teaching experience: calculus spring 2020
- Associate teaching experience: high numerical analysis autumn 2019

Seminar and Conference Talks

- The 631st Doctoral Student Forum of Tsinghua University April 2024